IRON WITHIN RESEARCH

# SS-31 — Certificates of Analysis

Every batch we have tested for this product, newest first.
Match the strength on your vial to the rows below and turn to that page.


| PAGE | STRENGTH | BATCH / LOT |
|---|---|---|
| **2–3** | 22mg, 50mg | 10/9/2026 · Lot IWR-1010-SS31 |
| **4–5** | 21mg, 50mg | 9/13/2026 · Lot IWR-090926SS |
| **6–7** | 12mg, 50mg | 8/28/2026 · Lot IWR-082526SS |
| **8–9** | 12mg, 50mg | 8/13/2026 · Lot IWR-081026SS |
| **10–12** | 21mg, 50mg, 83mg | 8/2/2026 · Lot IWR-2607-SS50L |
| **13** | 50mg, 71mg | 6/8/2026 |


Page numbers include this index page. Testing is per production batch; one batch can cover several vial sizes.

SUMMIT
BIOANALYTICS

# Certificate of Analysis

**#SBA-26-0299**

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
**Iron Within Research**

SAMPLE
**SS-31 — 50 mg**

**PASS**

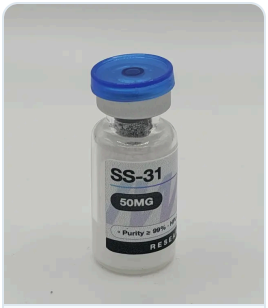
| BATCH # | PURITY | MASS | IDENTITY |
|---|---|---|---|
| IWR-1010-SS31 | 99.81% | 52.22 mg | Confirmed |

| | | | |
|---|---|---|---|
| COA #: | SBA-26-0299 | Method: | Full QC Panel |
| Lot Number: | IWR-1010-SS31 | Analysis Date: | 2026-10-09 |
| Labeled Content: | 50 mg | Date Received: | 2026-10-06 |
| | | Appearance: | White lyophilized powder, intact seal |

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

| Test | Specification | Result |
|---|---|---|
| Identity | SS-31 | Confirmed |
| Purity (HPLC) | >= 98.0% | 99.81% |
| Content (per vial) | Report only | 52.22 mg/vial |
| Fentanyl Screen (LC-MS/MS) | Not detected | Not detected |
| Endotoxins (LAL) | Report Result | 0.02 EU/mL |
| Sterility Screen (rapid PCR) | No growth | No growth |
| Heavy Metals (ICP-MS) | ICH Q3D limits | All analytes within limits |




SAMPLE AS RECEIVED

*Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.*

Kasey Malganius

**Kasey Malganius**
Lab Director
2026-10-09

| | |
|---|---|
| COA #: | **SBA-26-0299** |
| Access Code: | **6S4D7KF4** |
| Analyst: | **A. Joseph** |
| Verify: | summitbioanalytics.com/verify |
| Issued: | 2026-10-09 |

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

# Certificate of Analysis

**#SBA-26-0192**

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
**Iron Within Research**

SAMPLE
**SS-31 — 50 mg**

**PASS**

| BATCH # | PURITY | MASS | IDENTITY |
|---|---|---|---|
| IWR-090926SS | 99.82% | 54.21 mg | Confirmed |

| | | | |
|---|---|---|---|
| COA #: | SBA-26-0192 | Method: | Full QC Panel |
| Lot Number: | IWR-090926SS | Analysis Date: | 2026-09-13 |
| Labeled Content: | 50 mg | Date Received: | 2026-09-07 |
| | | Appearance: | White lyophilized powder, intact seal |
| | | Vial Cap: | Clear #A4A2A7 |

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

| Test | Specification | Result |
|---|---|---|
| Identity | SS-31 | Confirmed |
| Purity (HPLC) | >= 98.0% | 99.82% |
| Content (per vial) | Report only | 54.21 mg/vial |
| Fentanyl Screen (LC-MS/MS) | Not detected | Not detected |
| Endotoxins (LAL) | Report Result | 0.02 EU/mL |
| Sterility Screen (rapid PCR) | No growth | No growth |
| Heavy Metals (ICP-MS) | ICH Q3D limits | All analytes within limits |

*Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.*

Kasey Malganius

**Kasey Malganius**
Lab Director
2026-09-13

| | |
|---|---|
| COA #: | **SBA-26-0192** |
| Access Code: | **AWDGSSNW** |
| Analyst: | **R. Jeffereson** |
| Verify: | summitbioanalytics.com/verify |
| Issued: | 2026-09-13 |



This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

# Certificate of Analysis

**#SBA-26-0165**

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
**Iron Within Research**

SAMPLE
**SS-31 — 50 mg**

**PASS**

| BATCH # | PURITY | MASS | IDENTITY |
|---|---|---|---|
| IWR-082526SS | 99.13% | 51.12 mg | Confirmed |

| | | | |
|---|---|---|---|
| COA #: | SBA-26-0165 | Method: | Full QC Panel |
| Lot Number: | IWR-082526SS | Analysis Date: | 2026-08-28 |
| Labeled Content: | 50 mg | Date Received: | 2026-08-25 |
| | | Appearance: | White lyophilized powder, intact seal |
| | | Vial Cap: | Light blue #7FB2E5 |

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

| Test | Specification | Result |
|---|---|---|
| Identity | SS-31 | Confirmed |
| Purity (HPLC) | >= 98.0% | 99.13% |
| Content (per vial) | Report only | 51.12 mg/vial |
| Fentanyl Screen (LC-MS/MS) | Not detected | Not detected |
| Endotoxins (LAL) | Report Result | 0.04 EU/mL |
| Sterility Screen (rapid PCR) | No growth | No growth |
| Heavy Metals (ICP-MS) | ICH Q3D limits | All analytes within limits |

*Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.*

Kasey Malganius

**Kasey Malganius**
Lab Director
2026-08-28

| | |
|---|---|
| COA #: | **SBA-26-0165** |
| Access Code: | **X5APCVU3** |
| Analyst: | **K. Reyes** |
| Verify: | summitbioanalytics.com/verify |
| Issued: | 2026-08-28 |



This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
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# Certificate of Analysis

**#SBA-26-0082**

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
**Iron Within Research**

SAMPLE
**SS-31 — 50 mg**

**PASS**

| BATCH # | PURITY | MASS | IDENTITY |
|---|---|---|---|
| IWR-081026SS | 99.13% | 51.12 mg | Confirmed |

| | | | |
|---|---|---|---|
| COA #: | SBA-26-0082 | Method: | Full QC Panel |
| Lot Number: | IWR-081026SS | Analysis Date: | 2026-08-13 |
| Labeled Content: | 50 mg | Date Received: | 2026-08-10 |
| | | Appearance: | White lyophilized powder, intact seal |
| | | Vial Cap: | Light blue #7FB2E5 |

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

| Test | Specification | Result |
|---|---|---|
| Identity | SS-31 | Confirmed |
| Purity (HPLC) | >= 98.0% | 99.13% |
| Content (per vial) | Report only | 51.12 mg/vial |
| Fentanyl Screen (LC-MS/MS) | Not detected | Not detected |
| Endotoxins (LAL) | Report Result | 0.04 EU/mL |
| Sterility Screen (rapid PCR) | No growth | No growth |
| Heavy Metals (ICP-MS) | ICH Q3D limits | All analytes within limits |

*Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.*

Kasey Malganius

**Kasey Malganius**
Lab Director
2026-08-13

| | |
|---|---|
| COA #: | **SBA-26-0082** |
| Access Code: | **79J2HA45** |
| Analyst: | **A. Grinder** |
| Verify: | summitbioanalytics.com/verify |
| Issued: | 2026-08-13 |



This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

# Certificate of Analysis

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net




| | |
|---|---|
| **Client:** | Iron Within Research |
| **Accession #:** | 2606250659 |
| **Search Code:** | Iron2606250659 |
| **Received:** | 06/25/2026 |
| **Reported:** | 06/29/2026 |
| **Lot:** | IWR5506 |

## Sample Summary

| | | | |
|---|---|---|---|
| **Product:** | SS-31 50mg | **Purity:** | 99.81% |
| **Identity:** | Confirmed | **Net Content:** | 54.83 mg |
| **Appearance:** | White Lyophilized Powder | | |
| **Fentanyl Screen:** | Negative | | |

## Analytical Results

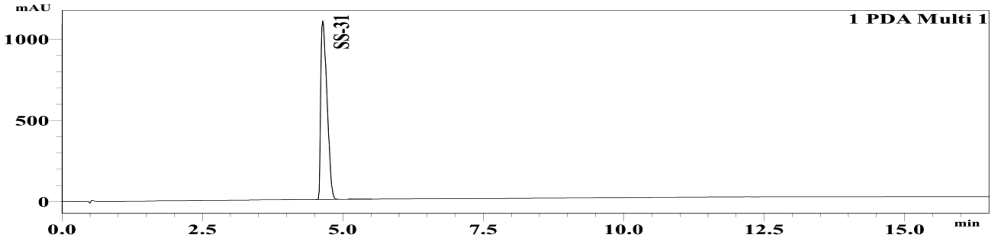
| Test | Result |
|---|---|
| **Identity (LC-MS)** | SS-31 |
| **Purity (HPLC-UV)** | 99.81% |
| **Net Content** | 54.83 mg |

Method: Endotoxin testing performed using Limulus Amebocyte Lysate assay in accordance with USP <85> under validated laboratory conditions.

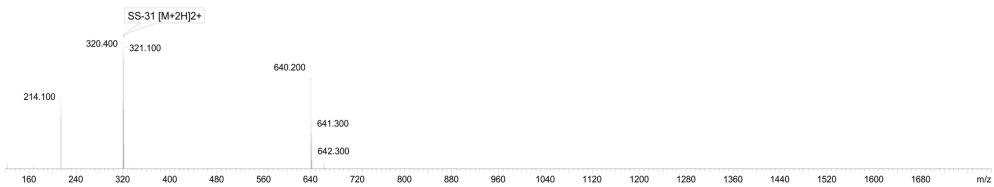
| | | |
|---|---|---|
| **Endotoxin Replicate 1:** | Pass | Assay Sensitivity: ≤0.05 EU/mL |
| **Endotoxin Replicate 2:** | Pass | Assay Sensitivity: ≤0.05 EU/mL |

Method: HPLC with UV detection coupled with mass spectrometry (LC-MS).

**Chromatogram**




**Mass Confirmation**




Principal Chemist

FreedomDiagnosticsTesting.com
Admin@FreedomDiagnostics.net
**Proudly Owned and Operated in the USA**

The peptide purity analysis reported here was conducted using LCMS/MS under standard laboratory conditions. This analysis is intended for informational purposes only and is specific to the sample (s) provided. The peptides tested are intended for research use only and are not approved for human or veterinary use, diagnostic, therapeutic, or clinical applications. Results should be interpreted by qualified professionals within the scope of the intended research. The accuracy and reliability of the test may be influenced by sample integrity, handling, and other experimental variables.

SUMMIT
BIOANALYTICS

# Certificate of Analysis

**#SBA-26-0058**

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
**Iron Within Research**

SAMPLE
**SS-31 — 50 mg**

**PASS**

| BATCH # | PURITY | MASS | IDENTITY |
|---|---|---|---|
| IWR-2607-SS50 | 99.29% | 53.21 mg | Confirmed |

| | | | |
|---|---|---|---|
| COA #: | SBA-26-0058 | Method: | Full QC Panel |
| Lot Number: | IWR-2607-SS50 | Analysis Date: | 2026-08-02 |
| Labeled Content: | 50 mg | Date Received: | 2026-07-29 |
| | | Appearance: | White lyophilized powder, intact seal |

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

VIAL CAP
**Pink**
#FB84B0

CLOSURE AS RECEIVED

| Test | Specification | Result |
|---|---|---|
| Identity | SS-31 | Confirmed |
| Purity (HPLC) | >= 98.0% | 99.29% |
| Content (gross) | Report only | 53.21 mg |
| Fentanyl Screen (LC-MS/MS) | Not detected | Not detected |
| Endotoxins (LAL) | Report Result | 0.10 EU/mL |

| Test | Specification | Result |
|---|---|---|
| **Sterility Screen (rapid PCR)** | No growth | **No growth** |
| **Heavy Metals (ICP-MS)** | ICH Q3D limits | **All analytes within limits** |

*Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.*



**Kasey Malganius**
Lab Director
2026-08-02

| | |
|---|---|
| COA #: | **SBA-26-0058** |
| Access Code: | **8Y5CXXAQ** |
| Analyst: | **R. Whitfield** |
| Verify: | summitbioanalytics.com/verify |
| Issued: | 2026-08-02 |



This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

FREEDOM DIAGNOSTICS

# Certificate of Analysis

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net




| | |
|---|---|
| **Client:** | Iron Within Research |
| **Accession #:** | 2606080224 |
| **Search Code:** | Iron2606080224 |
| **Received:** | 06/08/2026 |
| **Reported:** | 06/09/2026 |
| **Lot:** | IWR6526 |

## Sample Summary

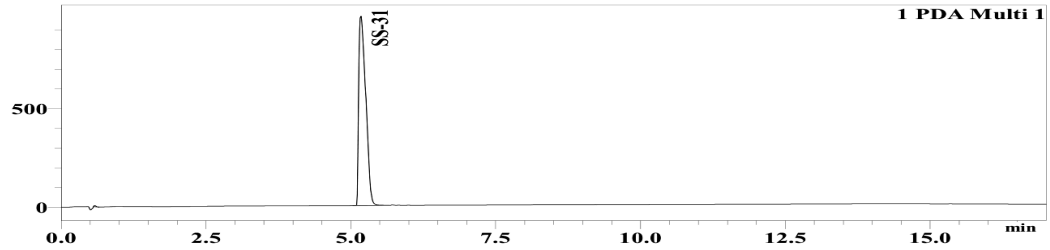
| | | | |
|---|---|---|---|
| **Product:** | SS-31 50mg | **Purity:** | 99.80% |
| **Identity:** | Confirmed | **Net Content:** | 51.71 mg |
| **Appearance:** | White Lyophilized Powder | **Fentanyl Screen:** | Negative |

## Analytical Results

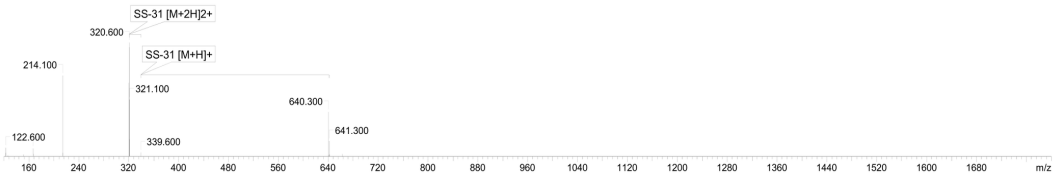
| Test | Result |
|---|---|
| **Identity (LC-MS)** | SS-31 |
| **Purity (HPLC-UV)** | 99.80% |
| **Net Content** | 51.71 mg |

Method: HPLC with UV detection coupled with mass spectrometry (LC-MS).

**Chromatogram**




**Mass Confirmation**




Principal Chemist

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net

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